Introduction to Artificial Intelligence

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Introduction: What Is Artificial Intelligence?

Artificial Intelligence (AI) is the branch of computer science that builds machines and software capable of performing tasks that normally require human intelligence. These tasks include understanding language, recognising images, making decisions, solving problems, translating speech and learning from experience. Instead of following a fixed list of instructions written line by line by a programmer, an AI system usually learns patterns from large amounts of data and then uses those patterns to make predictions or take actions.

Artificial Intelligence matters because it is now part of almost every industry, from healthcare and banking to agriculture, education and entertainment. Governments, companies and universities around the world treat AI as a general-purpose technology, similar in importance to electricity or the internet, because it can improve productivity and open up entirely new products and services.

History of Artificial Intelligence

The idea of thinking machines is old, but the modern field began in the 1950s. In 1950 the British mathematician Alan Turing published the paper "Computing Machinery and Intelligence", which asked whether machines can think and proposed the Imitation Game, now called the Turing Test. In the Turing Test a human judge chats with a hidden person and a hidden machine; if the judge cannot reliably tell which is which, the machine is said to show intelligent behaviour.

The term "Artificial Intelligence" was coined by John McCarthy, an American computer scientist. He used it in 1955 in a proposal for a summer research project, and the term became famous at the Dartmouth Workshop held at Dartmouth College in Hanover, New Hampshire, in the summer of 1956. The proposal was written by John McCarthy, Marvin Minsky, Nathaniel Rochester and Claude Shannon. Because of this workshop, 1956 is widely regarded as the birth year of AI as an academic field, and John McCarthy is often called one of the fathers of AI.

The field has seen cycles of excitement and disappointment. Periods of reduced funding and interest in the 1970s and again in the late 1980s are known as "AI winters". Progress returned with faster computers, huge datasets and better algorithms. In 1997 IBM's Deep Blue defeated the world chess champion Garry Kasparov. In 2012 a deep neural network called AlexNet won the ImageNet image recognition contest by a large margin, starting the deep learning era. In 2016 DeepMind's AlphaGo defeated the Go champion Lee Sedol. In 2017 researchers at Google introduced the Transformer architecture, and in November 2022 OpenAI released ChatGPT, which brought generative AI to hundreds of millions of ordinary users.

Types of AI: Narrow AI and General AI

AI is commonly divided into three levels based on capability. Narrow AI, also called weak AI or Artificial Narrow Intelligence (ANI), is designed to do one specific task or a small set of tasks. A spam filter, a face unlock system, a chess program, a voice assistant and a recommendation engine are all narrow AI. Every AI system in use today is narrow AI, even very powerful chatbots, because each is limited to what it was trained to do and does not truly understand the world like a person.

General AI, also called strong AI or Artificial General Intelligence (AGI), is a hypothetical system that could learn and perform any intellectual task that a human can, across many different areas, and transfer knowledge from one domain to another without being retrained. AGI does not yet exist. The key difference between narrow AI and general AI is therefore scope and flexibility: narrow AI is specialised and limited to its task, while general AI would be flexible, adaptable and human-level across tasks. A third, more speculative level is Artificial Superintelligence (ASI), an AI that would surpass the best human minds in nearly every field.

How AI Works: Main Approaches

Modern AI is built mainly on machine learning, in which algorithms learn from examples. Deep learning is a powerful form of machine learning that uses multi-layered neural networks. Older approaches include rule-based expert systems, where human experts write explicit if-then rules, and search algorithms, which explore possible moves to find the best one. Major sub-fields of AI include machine learning, natural language processing (understanding and generating human language), computer vision (understanding images and video), robotics, speech recognition and generative AI, which creates new text, images, audio and code.

Every AI project needs three essentials: data to learn from, an algorithm or model that learns patterns, and computing power to train and run the model. Graphics Processing Units (GPUs) are widely used because they can perform many calculations in parallel.

Everyday Uses of AI

People use AI many times a day without noticing. Three everyday uses of AI are voice assistants such as Google Assistant, Siri and Alexa that understand spoken commands; recommendation systems on YouTube, Netflix, Spotify and Amazon that suggest videos, songs and products; and navigation apps such as Google Maps that predict traffic and suggest the fastest route. Other common examples include email spam filters, face unlock on smartphones, language translation, autocorrect and predictive typing, fraud detection by banks, chatbots for customer service, and photo apps that automatically group pictures by person.

Beyond daily life, AI helps doctors detect diseases in X-rays and scans, helps farmers monitor crops with drone images, helps factories predict machine failures, helps banks approve loans, and helps teachers personalise lessons for students.

AI in India

India is investing heavily in AI. In March 2024 the Government of India approved the IndiaAI Mission with an outlay of more than Rs. 10,000 crore (about Rs. 10,372 crore) to build computing infrastructure, datasets, skills and startups. The NITI Aayog released the National Strategy for Artificial Intelligence in 2018 with the slogan "AI for All". Indian examples of AI in action include Bhashini, a national language translation platform that supports many Indian languages; AI-based crop advisories for farmers; AI chatbots used by banks and the railways; and AI tools used by hospitals for screening diabetic eye disease and tuberculosis from images. Indian IT companies such as HCL, TCS, Infosys and Wipro, along with many startups in Chennai and Bengaluru, build AI products for the world.

Challenges and the Future of AI

AI also brings challenges. Systems can be biased if their training data is biased, they can make confident mistakes, they raise privacy concerns because they need large amounts of data, and they may change the nature of jobs. Researchers and governments are working on responsible AI, regulation and safety. The future of AI is expected to include more capable assistants, AI agents that can complete multi-step tasks, AI in scientific discovery, and wider use on small devices at the edge. The long-term goal for many researchers remains artificial general intelligence, though experts disagree about when, or whether, it will arrive.

Quick Facts

- The term "Artificial Intelligence" was coined by John McCarthy in 1955 and popularised at the Dartmouth Workshop of 1956.
- Narrow AI performs specific tasks; general AI (AGI) would match human ability across all tasks and does not yet exist.
- Alan Turing proposed the Turing Test in his 1950 paper "Computing Machinery and Intelligence".
- Three everyday uses of AI: voice assistants, recommendation systems and navigation apps.
- India approved the IndiaAI Mission in March 2024 with an outlay of over Rs. 10,000 crore.